

DATA ANALYTICS PORTFOLIO

RIDE-BOOKING PERFORMANCE ANALYSIS

Excel Dataset + Power BI Dashboard

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1. Executive Summary

This report analyzes 150,000 ride-booking records from a full 2025 operating year, covering demand, cancellations, revenue, vehicle-type mix, geographic spread, and service quality. A Power BI dashboard translates that dataset into an operations-facing view built around five headline metrics: Completed Bookings, Lost Bookings, Revenue, Total Distance, and Average Distance, with drill-downs by month, vehicle type, and pickup/drop location.

Note on scope: the source material for this project is a ride-booking operations dataset (bookings, cancellations, vehicle types, fares, ratings) rather than a retail sales dataset. This report follows a standard consulting structure — executive summary, objectives, KPIs, trend and segment analysis, insights, recommendations — but every section below is built strictly from what the dataset actually contains. There is no profit, cost, or product-category data in this dataset, so those concepts are not fabricated; the equivalent, data-supported analyses (vehicle-type performance in place of product/category performance, and pickup/drop-location performance in place of regional performance) are used instead.

Overall, the business completed 93,000 of 150,000 requested bookings (a 62.0% completion rate), generating ₹51.85 million (~₹52M) in recorded booking value across roughly 2.51 million kilometres of completed and incomplete trips. The most significant finding is not on the revenue side but on the demand side: 57,000 bookings (38.0%) were never fulfilled — split fairly evenly between driver cancellations, customer cancellations, no-driver-found events, and incomplete trips — representing a substantial pool of unrealized revenue that is larger, in count, than any single vehicle type's completed-booking volume.

Auto leads every vehicle type on both booking volume and revenue; demand is broadly stable month to month rather than sharply seasonal; and service quality is solid, with average driver and customer ratings both above 4.2 out of 5. Section 14 documents two dashboard-construction issues (a missing vehicle-type lookup row and a duplicated icon mapping) and a data-integrity issue (non-unique booking IDs) that are worth fixing before this dashboard is presented externally.

2. Business Problem

A ride-booking platform generates a high volume of transactional data every day — requests, cancellations, completions, fares, distances, and ratings — but that data is only useful if it can be turned into a small number of monitorable metrics and a small number of clear questions: Is demand growing or shrinking? Which vehicle types and locations drive the business? How much demand is being lost to cancellations, and why? Is service quality holding up?

The Business Performance dashboard built for this project answers those questions in a single, filterable view. Rather than requiring a stakeholder to query 150,000 raw rows, the dashboard surfaces five

headline KPIs, a 12-month trend, a vehicle-type revenue breakdown, and top pickup/drop locations, all filterable by month and quarter — turning a raw operations log into a monitorable, decision-ready view.

3. Project Objectives

- Establish core operational KPIs: completed bookings, lost bookings, revenue, and trip distance.
- Track monthly booking and revenue trends to identify growth, decline, or seasonal patterns.
- Compare performance across the seven vehicle types offered on the platform.
- Quantify how much demand is lost to cancellations and incomplete trips, and why.
- Identify the highest-demand pickup and drop locations across the platform's 176-location network.
- Assess service quality through driver and customer ratings, and understand the payment method mix.
- Provide interactive Month/Quarter filtering so a stakeholder can drill into any period without rebuilding the report.

4. Dataset Overview

The workbook contains one primary data table (150,000 rows, 19 columns) plus two small lookup tables used to drive dashboard visuals: a vehicle-type-to-icon mapping (6 rows) and a booking-status-to-icon mapping (5 rows).

Field	Description
Date / Time	Booking date and time-of-day; the dataset spans Jan 1 – Dec 30, 2025 (364 of 365 days — see Section 5).
Booking ID / Customer ID	Unique identifiers for the booking and the rider (104,114 distinct customers across 150,000 bookings).
Booking Status	Completed, Cancelled by Customer, Cancelled by Driver, No Driver Found, or Incomplete.
Vehicle Type	One of 7 categories: Auto, Go Mini, Go Sedan, Bike, Premier Sedan, eBike, Uber XL.
Pickup / Drop Location	One of 176 named locations each, giving a granular view of geographic demand.
Booking Value / Ride Distance	Fare and trip distance; populated only where a trip actually occurred (see Section 5).
Driver / Customer Rating	1–5 scale, populated only for Completed bookings.
Payment Method	UPI, Cash, Uber Wallet, Credit Card, or Debit Card.
Cancellation / Incomplete Reason fields	Free-text-style categorical reason codes tied to each non-completed status.

This structure supports exactly the kind of analysis the dashboard performs: every booking can be sliced by when it happened, what vehicle was requested, where it started and ended, whether it succeeded, and — if it succeeded — how much it earned and how it was rated.

5. Data Preparation & Data Quality Review

Reviewing the dataset before analysis surfaced a consistent, well-structured pattern in how fields are populated by booking outcome, plus a few issues worth flagging:

- **Fields populate consistently with trip outcome** — Booking Value, Ride Distance, and Payment Method are populated for exactly 102,000 rows (Completed + Incomplete), while Driver Rating and Customer Rating are populated for exactly 93,000 rows (Completed only). This is internally consistent: a trip that never started can't have a fare, distance, or rating, and a trip that started but didn't finish (Incomplete) can still record a partial fare/distance but not a post-trip rating.
- **Booking ID is not a reliable unique key** — 1,233 rows share a Booking ID with another row (1,224 distinct IDs are duplicated), and the duplicate pairs checked represent genuinely different bookings on different dates with different customers, not accidental re-entry of the same trip. Any downstream analysis or join keyed on Booking ID should account for this rather than assuming uniqueness.
- **One calendar day is missing** — the dataset covers January 1 through December 30, 2025 (364 of 365 days); December 31 has no records. Immaterial for monthly analysis, but worth noting for anyone computing a strict full-year daily average.
- **No missing values in the core dimension fields** — Date, Booking ID, Customer ID, Booking Status, Vehicle Type, and both location fields have zero nulls across all 150,000 rows; the only nulls that exist are the outcome-dependent ones described above.

These checks were used to validate every KPI in Section 6 and every chart discussed in Sections 8–12 before drawing any conclusion from them.

6. Key Performance Indicators

The five headline metrics from the dashboard's summary cards, cross-checked directly against the underlying dataset:

COMPLETED BOOKINGS 93,000	LOST BOOKINGS 57,000	COMPLETION RATE 62.0%
REVENUE ₹51.85M	TOTAL DISTANCE 2.51M km	AVG. DISTANCE / TRIP 24.64 km

KPI	What It Measures	Why It Matters	Verified Result
Completed Bookings	Rides that were fully fulfilled	The platform's true delivered volume	93,000 (62.0% of all requests)
Lost Bookings	Cancelled (by either party), no-driver-found, or incomplete rides	Unrealized demand and a direct signal of supply or reliability gaps	57,000 (38.0% of all requests)
Revenue	Sum of Booking Value across Completed + Incomplete trips	Total recorded booking value for the year	₹51,846,183
Total Distance	Sum of Ride Distance across Completed + Incomplete trips	Total operational distance covered by the fleet	2,512,975 km
Avg. Distance	Mean Ride Distance per trip (Completed + Incomplete)	Typical trip length; informs pricing and driver allocation	24.64 km

Definitional note

Revenue, Total Distance, and Avg. Distance are calculated across Completed and Incomplete trips together (102,000 rows), not Completed trips alone — this was confirmed by matching the dashboard's exact figures against the dataset. It's a reasonable choice (an Incomplete trip still covers real distance and often charges a partial fare), but it means “Revenue” here is not strictly “revenue from successfully completed rides only” — worth stating explicitly in any stakeholder-facing version of this KPI.

7. Power BI Dashboard Overview

The dashboard opens on an Overview page anchored by the five KPI cards above, a 12-month Completed Bookings trend line, a 12-month Revenue bar chart, a Booking Value by Vehicle Type bar chart, top pickup/drop location tables, and driver/customer rating cards — all filterable by Month and Quarter, with a vehicle-type image strip for visual navigation.

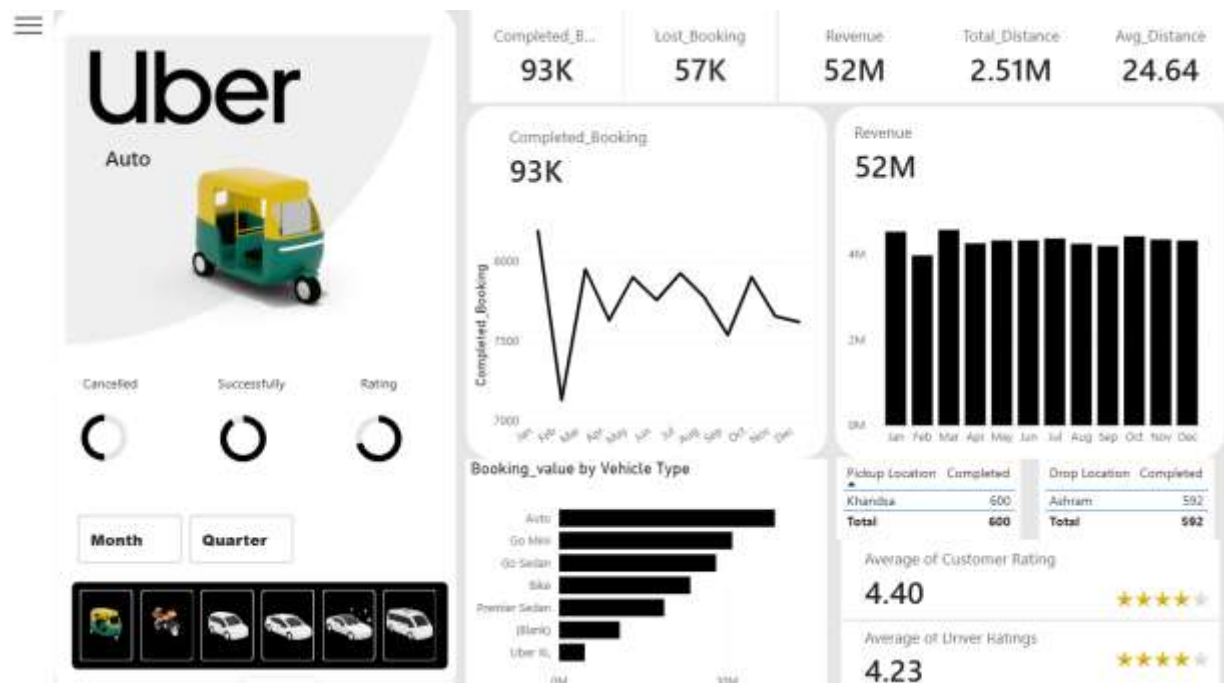


Figure 1 — Overview page, Power BI Desktop

Visualization	What It Shows	Business Question Answered
KPI Cards	Completed Bookings, Lost Bookings, Revenue, Total Distance, Avg. Distance	How is the platform performing right now, at a glance?
Completed_Booking trend line	Monthly completed booking volume, Jan–Dec	Is demand growing, shrinking, or stable month to month?
Revenue bar chart	Monthly revenue, Jan–Dec	Which months generate the most booking value?
Booking_value by Vehicle Type	Revenue contribution of each of the 7 vehicle types	Which vehicle types matter most to the business?
Pickup / Drop Location tables	Highest-volume pickup and drop points (Top 1 shown, sortable)	Where is demand concentrated geographically?
Average Customer / Driver Rating	Mean rating (1–5 scale) across all completed trips	Is service quality being maintained?
Cancelled / Successfully / Rating gauges	Intended as quick-glance completion and quality indicators	At-a-glance health check (see Section 14 — not rendering with data in the reviewed screenshot).
Month / Quarter slicers	Interactive filters	Let a stakeholder isolate any period without editing the report.

8. Booking & Revenue Trend Analysis

Monthly completed bookings are remarkably stable across the year, ranging narrowly between about 7,130 and 8,190 per month — there is no strong growth or decline trend, and no single month stands out as dramatically different from the rest. January is the strongest month for completed bookings (8,189); February is the weakest (7,130).

Month	Completed Bookings	Revenue
Jan	8,189	₹45.26L
Feb	7,130	₹39.71L
Mar	7,954	₹45.68L
Apr	7,632	₹42.54L
May	7,905	₹43.21L
Jun	7,757	₹43.26L
Jul	7,926	₹43.66L
Aug	7,780	₹42.44L
Sep	7,542	₹41.91L
Oct	7,905	₹44.17L
Nov	7,659	₹43.43L
Dec	7,621	₹43.20L

Revenue follows the same broadly flat pattern, peaking in March (₹45.68 lakh) narrowly ahead of January (₹45.26 lakh), and bottoming in February (₹39.71 lakh) — the same month with the fewest completed bookings. Since February is also the shortest calendar month, part of that dip is simply fewer days in the month rather than necessarily weaker daily demand; a per-day comparison would be needed to confirm whether February is genuinely soft or just short. Beyond that, the data does not show a clear seasonal pattern (e.g., a holiday-season spike), and no month should be assumed to be seasonal without more than one year of data to confirm the pattern repeats.

9. Vehicle Type Performance Analysis

In this dataset, vehicle type is the closest equivalent to a “product category” — it is the dimension that most directly drives both booking volume and revenue. Auto is the clear leader on both counts, generating close to a quarter of total platform revenue on its own:

Vehicle Type	Bookings (All Statuses)	Revenue	Share of Revenue
Auto	37,419	₹1,28,78,422	24.84%
Go Mini	29,806	₹1,03,38,496	19.94%

Vehicle Type	Bookings (All Statuses)	Revenue	Share of Revenue
Go Sedan	27,141	₹93,69,719	18.07%
Bike	22,517	₹78,37,697	15.12%
Premier Sedan	18,111	₹62,75,332	12.10%
eBike	10,557	₹36,18,485	6.98%
Uber XL	4,449	₹15,28,032	2.95%

Auto, Go Mini, and Go Sedan together account for nearly two-thirds of total revenue (62.85%), making them the platform's core business. Uber XL and eBike are the smallest contributors by a wide margin — together just under 10% of revenue — which may reflect genuinely lower demand for those categories, more limited vehicle availability, or simply a smaller addressable market (fewer riders need an XL vehicle or choose an eBike). The dashboard alone can't distinguish between a demand problem and a supply/availability problem here; that would need fleet-availability data this dataset doesn't include.

10. Cancellation & Unfulfilled Demand Analysis

This is the most consequential finding in the dataset. Of 150,000 total booking requests, only 93,000 (62.0%) were completed — the remaining 57,000 (38.0%) were lost to cancellation or non-completion, split across four distinct causes:

Outcome	Count	Share of All Requests
Cancelled by Driver	27,000	18.0%
Cancelled by Customer	10,500	7.0%
No Driver Found	10,500	7.0%
Incomplete	9,000	6.0%

Driver-initiated cancellations are the single largest loss category — at 18.0% of all requests, they are larger than customer cancellations and no-driver-found events combined. Breaking down the reason codes for each category shows no single dominant cause within any of them; each splits fairly evenly across 3–5 reasons, meaning this isn't a one-issue fix:

Cancelled by Driver (27,000 total)	Count
Customer related issue	6,837
The customer was coughing/sick	6,751
Personal & car related issues	6,726

Cancelled by Driver (27,000 total)	Count
More than permitted people in the car	6,686

Cancelled by Customer (10,500 total)	Count
Wrong address	2,362
Change of plans	2,353
Driver not moving towards pickup location	2,335
Driver asked to cancel	2,295
AC is not working	1,155

Notably, “Driver asked to cancel” appears as a customer-side cancellation reason (2,295 instances) — this represents cases where a driver-caused problem is being recorded against the customer's cancellation tally, which understates the true scale of driver-attributable lost bookings beyond the 27,000 already counted as direct driver cancellations.

11. Location Performance Analysis

With 176 distinct pickup and 176 distinct drop locations, this platform serves a wide, dispersed geographic network rather than concentrating demand in a handful of hubs. Across 93,000 completed bookings, the busiest single pickup location (Khandsa, 600 completed rides) is only about 14% above the per-location average of roughly 528 — a genuinely well-distributed demand pattern:

Top Pickup Locations	Completed	Top Drop Locations	Completed
Khandsa	600	Ashram	592
Barakhamba Road	594	Preet Vihar	589
Subhash Chowk	582	Sultanpur	584
Madipur	579	Noida Extension	579
Mehrauli	574	Narsinghpur / Dwarka Mor	574

Because demand is this evenly spread, there is no single “weak region” or “strong region” in the way a business with a handful of stores or territories would have — the more useful geographic question for this business is less “which location wins” and more “do any high-volume locations also have unusually high cancellation rates,” which would require joining the location and cancellation fields together, a deeper analysis this dashboard doesn't yet surface.

12. Ratings & Payment Method Analysis

Service quality, measured only on the 93,000 completed trips (ratings don't exist for non-completed bookings), is solid: average customer rating is 4.40 out of 5, and average driver rating is 4.23 out of 5. Both figures match the dashboard's cards exactly.

Payment method is recorded for 102,000 bookings (Completed + Incomplete). UPI dominates at 45.0% of transactions, roughly 1.8× the next-largest method (Cash, 24.9%) — a clear signal that digital-first payment infrastructure should be the platform's default assumption, not an alternative option:

Payment Method	Transactions	Share
UPI	45,909	45.01%
Cash	25,367	24.87%
Uber Wallet	12,276	12.04%
Credit Card	10,209	10.01%
Debit Card	8,239	8.08%

13. Key Business Insights

Insight 38.0% of all booking requests (57,000 of 150,000) are never completed, and driver-initiated cancellations alone (18.0%) exceed customer cancellations and no-driver-found events combined.

Business Impact This is the single largest gap between demand and delivered revenue in the dataset — larger in count than the completed-booking volume of every vehicle type except Auto and Go Mini combined.

Recommended Action Prioritize a driver-cancellation root-cause review before any demand-generation or marketing investment; closing even a fraction of this gap likely outweighs the revenue upside of most other initiatives in this report.

Insight Auto, Go Mini, and Go Sedan account for 62.85% of total revenue, while Uber XL and eBike together contribute under 10%.

Business Impact The business is heavily concentrated in three of seven vehicle types; a disruption to Auto specifically (24.84% of revenue alone) would be a material revenue event.

Recommended Action Investigate whether Uber XL and eBike's low share reflects genuine low demand or limited vehicle availability before deciding whether to invest in or deprioritize those categories.

Insight Monthly completed bookings and revenue are both flat within a narrow band all year (no month varies more than ~13% from the annual average), with February the softest month for both metrics.

Business Impact There is no evidence of a growth trend or a strong seasonal cycle in this single year of data — the business is stable but not currently expanding organically.

Recommended Action Treat February as worth a closer look (accounting for its shorter length) rather than assuming seasonality; if a genuine dip persists after adjusting for day count, consider a targeted February promotion.

Insight UPI accounts for 45.0% of all payments, nearly double the next-largest method (Cash, 24.9%), while card-based payments (Credit + Debit) together are just 18.1%.

Business Impact The customer base is strongly digital-payment-oriented; any friction in the UPI payment flow would affect a disproportionate share of transactions.

Recommended Action Ensure UPI payment reliability is treated as a top-priority technical concern, and evaluate whether card payment's smaller share reflects a UX gap worth addressing or simply genuine low preference.

Insight Demand is spread across 176 pickup and 176 drop locations with no single location exceeding roughly 14% above the per-location average.

Business Impact There is no over-concentration risk in any one location, but this also means location-specific interventions will have limited platform-wide impact.

Recommended Action Focus operational improvements (driver supply, cancellation reduction) on platform-wide policies rather than location-specific fixes, since no location is a disproportionate driver of the business.

Insight “Driver asked to cancel” is recorded as a customer-cancellation reason (2,295 instances, 21.9% of all customer cancellations) rather than as a driver cancellation.

Business Impact The true scale of driver-attributable lost bookings is likely understated by the 27,000 figure reported under “Cancelled by Driver,” since some driver-caused cancellations are being logged against the customer instead.

Recommended Action Review how this reason code is classified in the source system; if these are genuinely driver-initiated, reclassifying them would give a more accurate picture of driver-side cancellation volume.

Insight Average customer rating (4.40) is meaningfully higher than average driver rating (4.23) on the same set of completed trips.

Business Impact Customers are rating their experience more favorably than drivers are being rated in return — a gap worth understanding, since it could reflect genuine driver-side service issues or a stricter customer-rating culture on the platform.

Recommended Action Segment driver ratings by vehicle type and cancellation history to see whether the lower average is concentrated among specific driver segments rather than spread evenly.

Insight Incomplete trips (9,000, 6.0% of all requests) still record partial Booking Value and Ride Distance, contributing roughly ₹45.9 lakh to total reported “Revenue.”

Business Impact The headline Revenue KPI is not purely “revenue from successful trips” — it includes partial value from trips that did not finish, which could overstate realized revenue if read at face value by a stakeholder.

Recommended Action Consider adding a “Revenue (Completed only)” metric alongside the current definition so stakeholders can see both the inclusive and completed-only figures.

14. Data Quality & Dashboard Notes

Three issues were identified that should be resolved before this dashboard is shared externally or used to drive decisions:

- **eBike is missing from the vehicle-image lookup table** — the workbook's Veh_IMG sheet, which maps each Vehicle Type to an icon image for dashboard visuals, lists only 6 of the 7 vehicle types (Auto, Go Mini, Go Sedan, Bike, Premier Sedan, Uber XL) — eBike has no row. On the dashboard's “Booking_value by Vehicle Type” chart, the category that should read “eBike” instead displays as “(Blank)”, positioned exactly where eBike's revenue value (₹36.18 lakh) would sort. Adding an eBike row to Veh_IMG should resolve this.
- **Duplicate icon mapping for Booking Status** — the workbook's status-to-icon lookup table assigns the identical icon image to “Completed” and “Cancelled by Driver,” and a second identical icon to “Cancelled by Customer” and “Incomplete.” Any dashboard visual using this table to display a status icon would currently show a “cancel”-style icon next to “Completed” bookings. Each of the five statuses should have its own distinct icon.
- **Booking ID is not unique** — as noted in Section 5, 1,233 rows share a Booking ID with a different, genuinely distinct booking. If this field is ever used as a join key or a primary key in a downstream system, de-duplication logic (similar to the EmpID pattern used in this portfolio's HR Analytics project) would be needed first.

On the reviewed screenshot

The Cancelled, Successfully, and Rating gauge visuals on the dashboard's left panel rendered as empty outline circles with no data in the screenshot provided for this review. This may simply reflect the screenshot being captured mid-load or mid-navigation (a page-switch tooltip is visible in the same image) rather than a genuine data issue — it's noted here for completeness but not treated as a confirmed defect the way the two lookup-table issues above are.

15. Business Recommendations

- Launch a driver-cancellation reduction initiative first: at 18.0% of all requests, this is the single largest, most concentrated source of lost bookings in the dataset.
- Audit the “Driver asked to cancel” customer-cancellation reason code, since it likely masks additional driver-attributable cancellations beyond the reported 27,000.
- Protect and invest in UPI payment reliability given its 45% share of transactions, and investigate whether the lower card-payment share reflects a fixable friction point.
- Investigate the demand vs. availability question for Uber XL and eBike before making any investment or deprioritization decision on those vehicle types.
- Fix the two dashboard lookup-table issues (missing eBike row, duplicated status icons) and correct the Booking ID uniqueness issue before this report or dashboard is used in an external-facing or portfolio-review context.
- Add a “Revenue (Completed only)” KPI alongside the current inclusive Revenue figure so stakeholders aren't misreading partial-trip value as fully realized revenue.
- Investigate the driver-vs-customer rating gap (4.23 vs. 4.40) by vehicle type and cancellation history to identify whether it's concentrated in specific driver segments.

16. Conclusion

The platform's core business is healthy and stable: revenue and completed-booking volume hold steady across all twelve months, service quality ratings sit comfortably above 4.2 out of 5, and demand is spread broadly and evenly across a 176-location network rather than concentrated in a few fragile hubs. The clearest opportunity — and the clearest risk, if left unaddressed — is the 38.0% of all booking requests that never convert into a completed trip, with driver-initiated cancellations as the largest single contributor.

As a dashboard, this report demonstrates how a large (150,000-row) transactional log can be distilled into a small set of monitorable KPIs and trend views that let a stakeholder answer “how is the business doing?” in seconds rather than hours — provided the underlying lookup tables and identifiers are cleaned up as described in Section 14. With those fixes applied, the dashboard is well-positioned to support ongoing, faster, data-driven operational decisions.

17. Portfolio Project Summary

Project Name	Ride-Booking Performance Analysis (Business Performance 360° Dashboard)
Tools Used	Microsoft Excel, Power BI
Business Objective	Turn a 150,000-row ride-booking log into a monitorable set of KPIs and trend views covering demand, cancellations, revenue, vehicle-type mix, geography, and service quality.
Key Analysis	Data cleaning and structural review of a 19-column transactional dataset; KPI reconciliation against the Power BI dashboard; monthly trend, vehicle-type, cancellation-cause, location, and rating/payment analysis.
Key Findings	62.0% completion rate with 38.0% of demand unfulfilled (driver cancellations the largest single cause); Auto/Go Mini/Go Sedan drive 62.85% of revenue; demand is stable year-round and evenly spread across 176 locations; two dashboard lookup-table defects and a non-unique Booking ID field identified and documented.
Business Value	Gives operations stakeholders a single, filterable view to monitor completion rates, revenue, and service quality, and pinpoints driver-cancellation reduction as the highest-leverage opportunity in the dataset.

18. Skills Demonstrated

- Microsoft Excel: large-dataset review (150,000 rows), null-pattern analysis, duplicate-key detection, lookup-table auditing.
- Power BI: KPI card design, trend and bar visualizations, interactive Month/Quarter slicers, image-based category navigation.
- Data Preparation: verifying field population patterns against business logic, identifying non-unique identifiers, and confirming date-range completeness.
- Business Analysis: translating a transactional log into decision-ready KPIs, distinguishing genuine trends from artifacts (e.g., February's shorter length), and identifying the highest-leverage business opportunity in a dataset.
- Data Storytelling: structuring findings as Insight / Business Impact / Recommended Action so every finding connects to a concrete next step.
- Data-Driven Decision Making and Interactive Reporting: building a dashboard-and-report pairing designed to be audited against each other, not taken at face value.